

Pradyumna Sadhu

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EDUCATION

• University of California, Riverside

Masters and Ph.D. in Astronomy

Sept 2022 - Present

Riverside, CA, USA

◦ Advisor: Prof. Laura Sales

◦ CGPA: 4.0/4.0

◦ Dean's Distinguished Fellowship recipient

• Indian Institute of Technology, Hyderabad

Bachelor of Technology with Honors in Engineering Physics

Aug 2018 - May 2022

Hyderabad, India

◦ CGPA: 9.5/10.0

◦ Silver Medal for highest CGPA in the graduating class

◦ Research Advisors: Prof. Shantanu Desai (IIT Hyderabad, India), Dr. Yong Tian (NCU, Taiwan)

◦ Honors in Physics which included additional research projects and advanced courses

PUBLICATIONS

Three Key Publications:

1. **Pradyumna Sadhu**, Laura V. Sales, Julio F. Navarro, Raphaël Errani, Jose Benavides, Eric W. Peng.
The abundance and radial distribution of faint and ultra-faint dwarfs in galaxy clusters.
MNRAS, Accepted. DOI: 10.1093/mnras/stag1604.
2. **Pradyumna Sadhu**, Yong Tian (2024).
The Baryonic Faber Jackson Relation in Galaxy Groups.
MNRAS, Vol. 528, Issue 4, pp. 5612–5623. DOI: 10.1093/mnras/stae343
3. **Pradyumna Sadhu**, Sajal Gupta, Sowmya Seeram, Shantanu Desai (2021).
Yet another test of Radial Acceleration Relation for galaxy clusters.
Physics of the Dark Universe, Vol. 31. DOI: 10.1016/j.dark.2020.100765

In-prep Publication:

1. **Pradyumna Sadhu**, Laura V. Sales, Julio F. Navarro, Jens Stucker, Kaixiang Wang, Eric W. Peng.
Envelope UCDs as the observed tidal remnants of nucleated dwarf galaxies.

Other Publications:

1. **Pradyumna Sadhu**, Shantanu Desai (2022).
A test of galaxy cluster fundamental plane for the X-COP sample.
Journal of Cosmology and Astroparticle Physics, 2022. DOI: 10.1088/1475-7516/2022/01/058
2. **Pradyumna Sadhu**, Shantanu Desai (2022).
Characterization of the GRB prompt fundamental plane using Fermi-GBM data.
Journal of High Energy Astrophysics, Vol. 35, pp. 77–82. DOI: 10.1016/j.jheap.2022.06.003
3. **Pradyumna Sadhu**, Shantanu Desai (2021).
A test of Radial Acceleration Relation for the Giles et al Chandra cluster sample.
Physics of the Dark Universe, Vol. 33. DOI: 10.1016/j.dark.2021.100854
4. Jacobson-Galán, W.V. et al. (including **Pradyumna Sadhu**) (2023).
SN 2023ixf in Messier 101: Photo-ionization of Dense, Close-in Circumstellar Material in a Nearby Type II Supernova.
The Astrophysical Journal, Vol. 954. DOI: 10.3847/2041-8213/acf2ec.
Contributed as a co-observer at Lick Observatory when this close and bright supernova went off.

TALKS AND CONFERENCE PRESENTATIONS

- **248th meeting of the American Astronomical Society** *June 2026*
Pasadena, CA, USA
- **GALFRESCA 2025** *Sept 2025*
UC San Diego, CA, USA
- **Astronomical Society of India Meeting 2025** *Feb 2025*
NIT Rourkela, India
- **Seminar on Ultra-faint Dwarf Satellite Population in Galaxy Clusters within Λ CDM** *Feb 2025*
IIT Hyderabad, India (Invited)
- **Physics and Astronomy Student Seminar** *Sept 2024*
UC Riverside, CA, USA
- **SoCalDM workshop** *May 2024*
UC Riverside, CA, USA
- **Innovation in Appalachia: Undergraduate Symposium** *Aug 2021*
West Virginia University (online due to COVID-19)
- **Presision 2021** *Sept 2021*
Presidency University, Kolkata, India (online due to COVID-19)

CONFERENCE POSTER PRESENTATIONS

- **IAU Symposium 403: The Hidden Beauty of the Galactic Outskirts** *Oct 2025*
University of Cordoba, Cordoba, Spain
- **Santa Cruz Galaxies Workshop** *Aug 2025*
UC Santa Cruz, CA, USA
- **Dwarf Galaxies, Star Clusters, and Streams in the LSST Era Workshop** *July 2024*
University of Chicago, IL, USA
- **Astronomical Society of India Meeting 2022** *Feb 2022*
Online due to COVID-19
- **ESO Cluster Mass 2020** *Sept 2021*
European Southern Observatory (online due to COVID-19)

UNDERGRADUATE MENTORING

- **Sri Devaki Meduri (IIT Hyderabad, India)** *May 2026 - present*
The Faber-Jackson relation across mass scales in the IllustrisTNG simulation suite
- **Angela Claibourn (University of Hawai'i - Hilo, Hawai'i, USA)** *June 2025 - Dec 2025*
Time evolution of the stellar distributions and kinematics in satellite galaxies in TNG50

TECHNICAL SKILLS

- **Programming & Computing:** Python (NumPy, SciPy, AstroPy, GalPy, pandas, matplotlib, object oriented programming, parallel computing), C, HPCC
- **Astrophysical Simulation & Analysis Codes:** GADGET-2/4, AREPO, SMUGGLE, GALFIT, SUBFIND
- **Statistical Methods:** Bayesian analysis, data visualization
- **Tools & Writing:** Git/GitHub, Linux/Unix, ssh, LaTeX, scientific writing

HONORS AND AWARDS

- **GSA Travel Award** 2024, 2025, 2026
University of California, Riverside
- **Dean's Distinguished Fellowship** 2022
University of California, Riverside
- **Silver Medal** 2022
Indian Institute of Technology, Hyderabad
- **Research Excellence Award** 2022
Indian Institute of Technology, Hyderabad
- **Academic Excellence Award** 2018, 2020
Indian Institute of Technology, Hyderabad

TEACHING EXPERIENCE

- **Teaching Assistant for PHYS020 Exploring the Universe: an adventure in Astronomy** 2025 - present
University of California, Riverside
 - Holding discussion sections consisting a 30 minute lecture and a 10 minute quiz to create interest in undergraduate students across different disciplines for astronomy.
- **Teaching Assistant for Physics Laboratories** 2023 - 2024
University of California, Riverside
 - Taught PHYS 02LA, 02LB, and 02LC laboratory courses for pre-medical students by guiding students through experimental procedures and data analysis
 - Evaluated student performance and providing constructive feedback
- **Teaching Assistant for Special Relativity and Mathematical Physics** 2021 - 2022
Indian Institute of Technology, Hyderabad
 - Created and graded problem sets; evaluated the finals, for sophomore students
- **Online Math and Physics Tutor** 2022
Akshardaan and National Service Scheme, IIT Hyderabad
 - Provided educational support to economically limited high school students by teaching mathematics and physics online during COVID-19 pandemic
- **STEM Summer School Instructor** June 2024
University of California, Riverside
 - Assisted in teaching data analysis using Python to more than 50 high school students by developing hands-on coding exercises

LEADERSHIP AND OUTREACH

- **Co-President, Astronomy Union for Reading, Outreach and Research Activities** 2025 - Present
University of California, Riverside
 - Organizing astronomy seminars, journal clubs and public engagement events
- **Seminar Host for Physics and Astronomy Student Seminars** 2024 - Present
University of California, Riverside 
 - Assisting with organization and hosting of regular student seminars to facilitate knowledge sharing among graduate students
- **Public Telescope Observations** 2023 - Present
University of California, Riverside

- Volunteering at multiple public telescope observation sessions to educate community members about celestial objects and astronomical phenomena
- **Judge, Riverside Unified Science and Engineering Fair** Jan 2024
Riverside Unified School District
 - Evaluated student science projects and provided feedback to encourage young minds in scientific inquiry
- **Physics Content Developer** May 2020 - June 2020
Embibe 
 - Developed physics educational content for online learning platform for Joint Entrance Exam, one of the most competitive exams in the world
- **Founder and Treasurer, Darbar** 2024 - 2026
University of California, Riverside
 - Established student organization for promotion of Indian classical and folk art forms to build community and cultural awareness on campus
- **General Secretary, SPIC MACAY IIT-H Subchapter** 2020 - 2022
Indian Institute of Technology, Hyderabad 
 - Organized cultural events, concerts, and workshops by managing a team of volunteers and coordinating with a national organization
- **Editor, CosmicVarta** 2023
Indian Astronomy Community Outreach Platform 
 - Edited and curated Indian astronomy research content for public dissemination to bridge the gap between research community and general public